

Name	Subject Class	Class	Candidate Number
	2BI		



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2015

BIOLOGY

HIGHER 2

9648/02
24 AUGUST 2015
2 hours

Paper 2 Core Paper

Additional Material: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, index number and class on this answer booklet.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, circle the number of the Section B question you have answered in the grid opposite.

Fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
8	
Section B	
9 or 10	
Total	100

- 1 (a) Fig. 1.1 is a schematic diagram showing the transport pathways of extracellular and intracellular materials for digestion in a mammalian cell. Depending on the types of digested material, three possible pathways are initiated to deliver these materials for digestion within lysosomes, of which two are labelled as **A** and **B**.

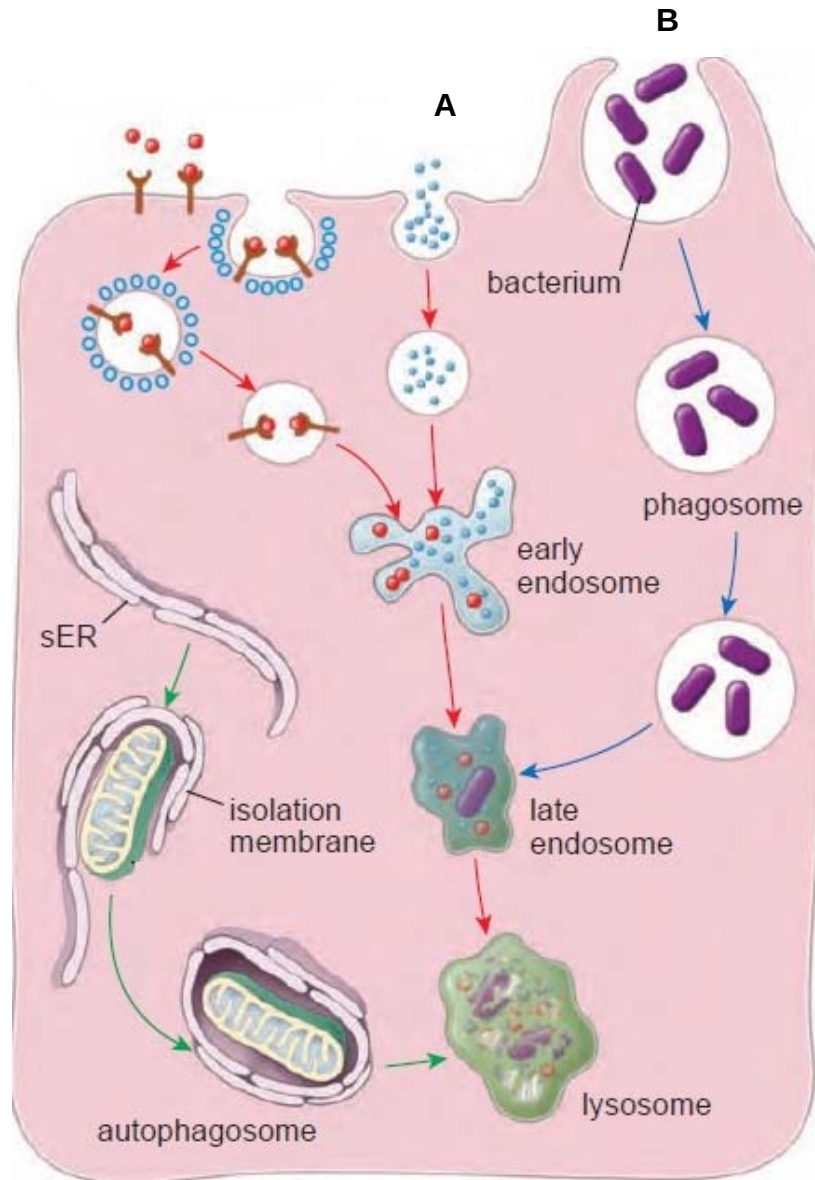


Fig. 1.1

- (i) State **one** property of the plasma membrane and explain how it enables process **A** to be carried out by a cell. [2]

- (ii) Degradation of worn out organelles such as mitochondria occurs inside most cells via autophagy. With reference to Fig. 1.1, describe the process of autophagy. [3]

- (iii) Process **B** requires rapid rearrangement of the cytoskeleton when extracellular materials bind to cell surface membrane receptors. This rapid rearrangement requires the activation of a variety of protein kinases which occurs when there is a surge in the cytosolic levels of calcium ions. Explain the role of calcium ions to bring about this cellular response. [3]

- (b) Lysosomes are rich in enzymes such as proteases and nucleases. The lysosomal membrane contains an unusually large amount of integral proteins, which are highly glycosylated with oligosaccharides on the surface of membrane facing the interior.

- (i) Briefly describe how an oligosaccharide is formed from glucose. [3]

(ii) Suggest the role of these glycosylated proteins in lysosomes.

[1]

[Total: 12m]

- 2 Transglutaminase (TGase) secreted by *Streptomyces mobaraense* is a valuable enzyme for the food industry. TGase, better known to chefs as “meat glue”, has the ability to bond protein-containing foods together. Some of its uses include the production of restructured meat and the hardening of fish protein paste.

TGase consists of 331 amino acids, including only one cysteine residue. The three amino acid residues at the catalytic site are cysteine 64, aspartate 255 and histidine 274, where the numbers denote the positions of the amino acids in the amino acid sequence of TGase. Fig. 2.1 shows the structure of TGase.

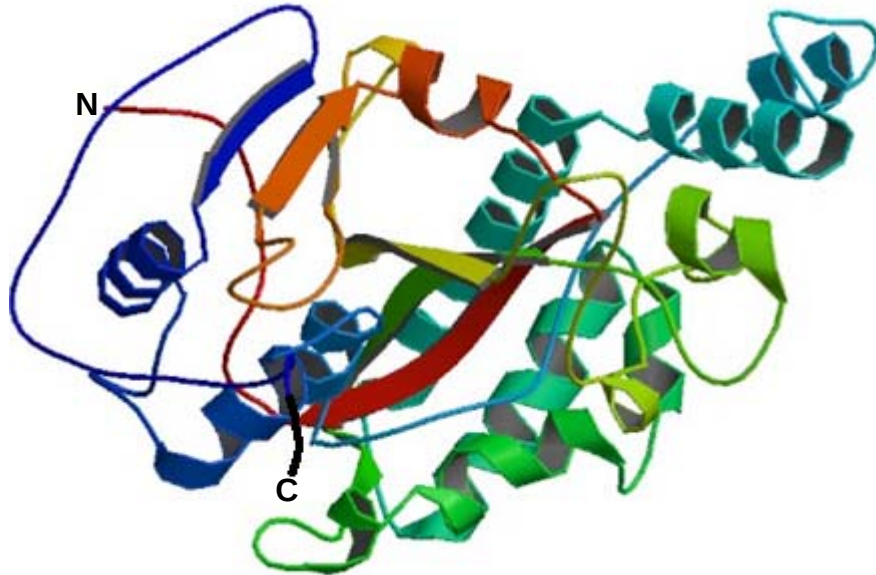


Fig. 2.1

- (a) With reference to Fig. 2.1 and your knowledge of haemoglobin, state **two** structural differences between TGase and haemoglobin. [2]

- (b) Explain what determines the precise position of the three catalytic residues in the active site of TGase. [2]

(c) Describe the mode of action of TGase.

[3]

[Total: 7m]

- 3 Fig. 3.1 shows the sequence of nucleotide bases in a section of a viral gene coding for the first seven amino acids of a viral protein, X.

CTTCTGTTTCTGCTGGAACAT

Fig. 3.1

- (a) Methionine is coded by the initiation codon. List the corresponding codons on the mRNA (in the 5' to 3' direction) transcribed from the template in Fig. 3.1. [1]

- (b) Protein X was hydrolyzed to release the first seven amino acids which comprises of four different amino acids. The number of each type of amino acid obtained is shown in Table 3.1.

Table 3.1

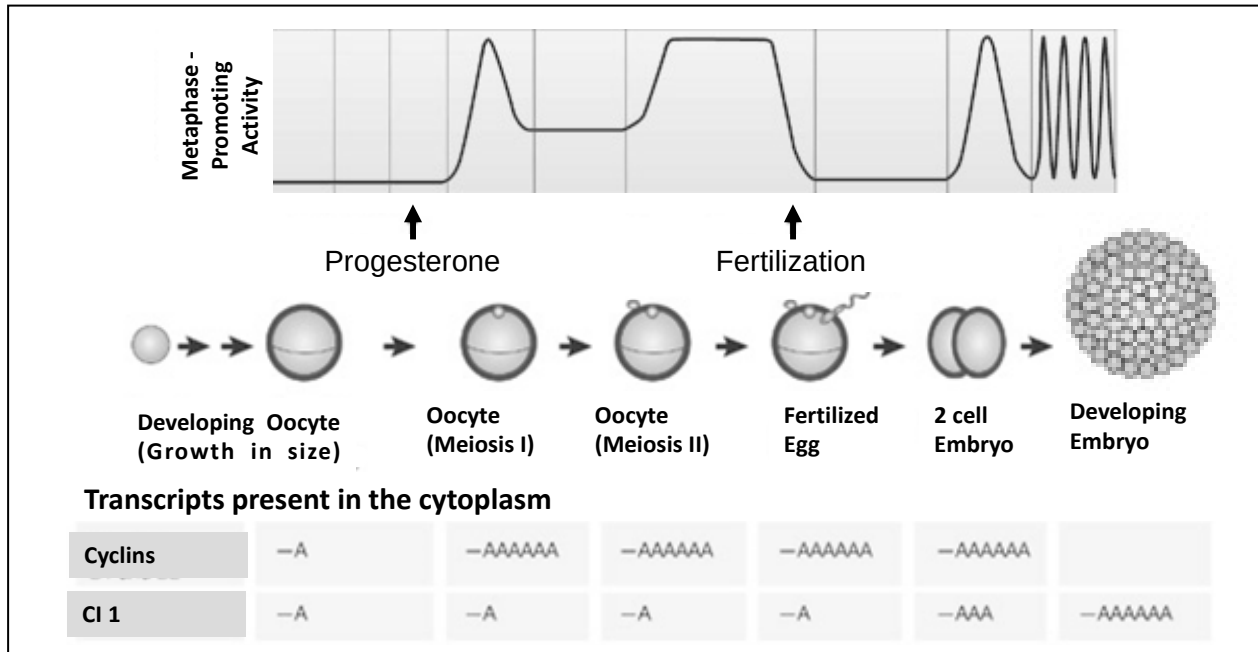
Amino acid	Number present
Phe	1
Met	1
Lys	2
Gln	3

Using information in Table 3.1, list the first seven amino acids of protein X in order, starting from the amino acid at the N-terminus. [1]

- (c) Explain if the replication of the influenza virus genome can be described as semi-conservative. [3]

[Total: 5m]

- 4 Fig. 4.1 shows a series of events that occur during the maturation of an oocyte leading to its fertilization and the development of the subsequent embryo. The corresponding metaphase-promoting activity due to proteins present in the various stages of the oocyte and the embryo are shown in a graph above the cells. The mRNA transcripts of cyclins and Cl 1 with their corresponding length of poly-A tails at various stages are shown below the cells.



Legend	
A	Short poly-A tail
AAA	Mid length poly-A tail
AAAAAAA	Long poly-A tail

Fig. 4.1

- (a) With reference to Fig. 4.1, state and explain the mode of control of gene expression that leads to the maturation of the oocyte until meiosis II. [4]

- (b)** Suggest the advantage of the type of gene control that is present in the early stages of the embryo. [2]

- (c)** Cyclin is a protein that can be found in most cells in the body. Explain how a mutation in the gene coding for cyclin could contribute to the development of cancer. [4]

[Total: 10m]

- 5 (a) Recent findings suggest that the mucus lining the human digestive tract is loaded with bacteriophages. These bacteriophages contain proteins which bind to the carbohydrate residues of mucoproteins as shown in Fig. 5.1. It has been proposed that these bacteriophages can play an important role as part of the body's defense system.

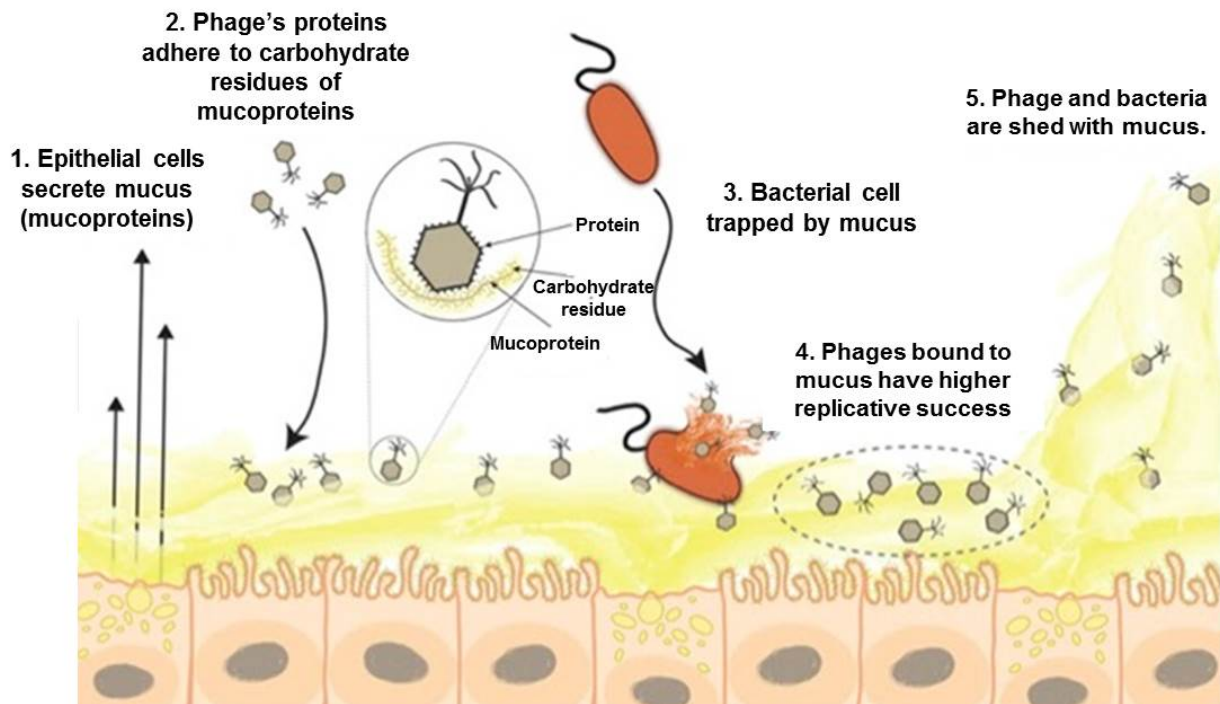


Fig. 5.1

- (i) Describe the life cycle of the bacteriophages shown in Fig. 5.1.

[4]

- (ii) With reference to Fig. 5.1, suggest how the way in which bacteriophages bind to mucoproteins is significant to their higher replicative success.

[2]

- (iii) Many other viruses may also be found within the human body e.g. the Human Immunodeficiency Virus, HIV. Compare between the lysogenic life cycle of a temperate bacteriophage and life cycle of a HIV. [3]

- (b) In bacteria, genes coding for enzymes involved in the same metabolic pathway are arranged into operons. Fig. 5.2 shows the changes in the activity of enzymes that synthesise tryptophan and utilise lactose in a cell after the addition of tryptophan and lactose.

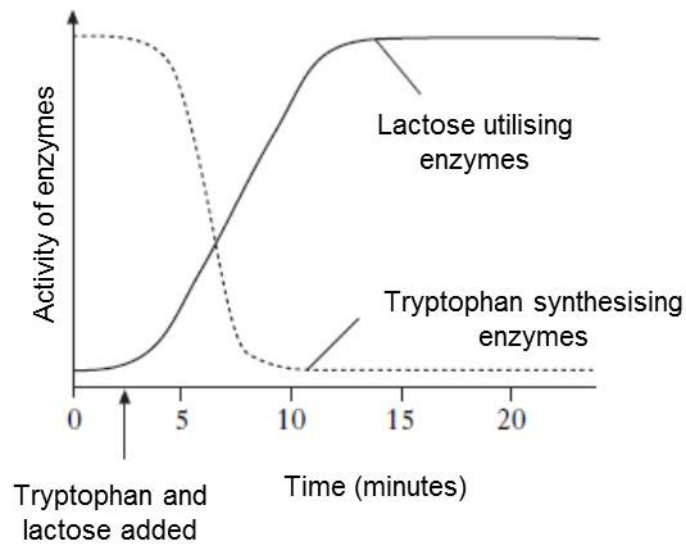


Fig. 5.2

A scientist altered the *trp* operator sequence *in vitro* such that it was no longer complementary to the binding sites of repressor proteins. The scientist also altered the *lac* repressor gene sequence *in vitro* such that the repressor binding site was no longer complementary to its inducer.

Describe and explain the effects of the mutations on the activity of lactose utilising enzymes and tryptophan synthesizing enzymes upon addition of lactose and tryptophan. [4]

[Total: 13m]

- 6 Analysis of genes on separate loci in tomato plants were carried out. In the experiment, pure-breeding tall tomato plants with smooth fruits were crossed with pure-breeding dwarf tomato plants with hairy fruits. All the plants in the F_1 generation were tall with smooth fruits.

When a F_1 plant was subjected to a test cross, the following offspring phenotypes were observed:

Tall with smooth fruit	80
Tall with hairy fruit	20
Dwarf with smooth fruit	19
Dwarf with hairy fruit	81

- (a) Using the symbols **T** and **t** for height and **H** and **h** for fruit type, draw a genetic diagram to explain the observed results of the test cross. [5]

(b) (i) State the expected dihybrid test cross ratio.

[1]

(ii) Explain why the observed test cross results deviate from the ratio stated in **(b)(i)**.

[3]

A chi-squared test was used to find out if there is any significant difference between the observed and expected ratio.

The formula for χ^2 and the table of probabilities are given as follows:

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad \nu = c - 1$$

where Σ = 'sum of...'

O = observed 'value'

E = expected 'value'

ν = degrees of freedom

c = number of classes

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

(c) Calculate the χ^2 value and conclude the results of the chi-squared test at 5% level of significance.

[3]

[Total: 12m]

- 7 (a) Fig. 7.1 is an electron micrograph of a mitochondrion from an animal cell.

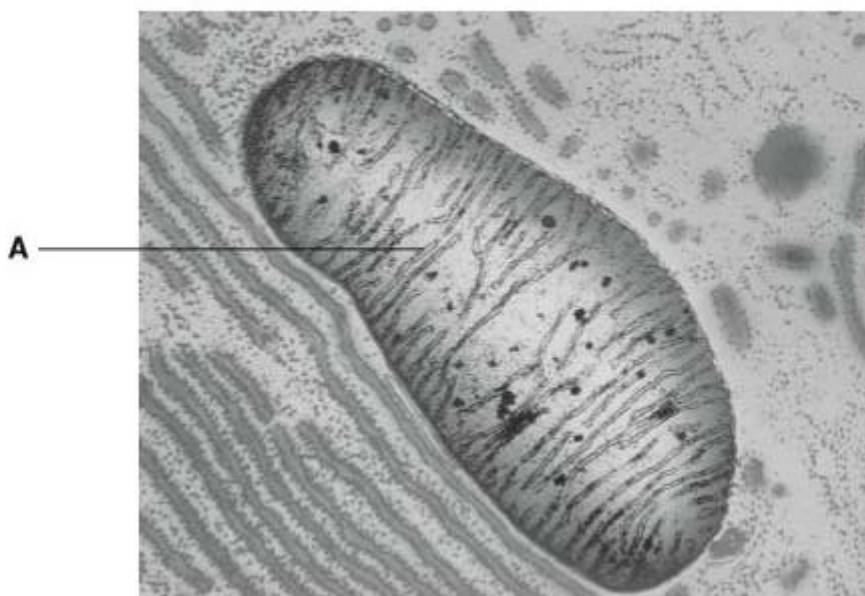


Fig. 7.1

- (i) Name the specific process that is carried out by structure **A** in the mitochondrion. [1]

- (ii) State **three** differences between the process of ATP formation in structure **A** and in the cytoplasm. [3]

- (iii) Suggest how the structure of a mitochondrion from a muscle cell would differ from that of a mitochondrion from a skin cell. [1]

- (b) The bacterium, *Halobacterium salinarium* (*H. salinarium*) utilises light energy to drive the production of ATP in the cell membrane. Fig. 7.2 shows how the protein, bacteriorhodopsin, present in the cell membrane of *H. salinarium* helps with the process of producing ATP.

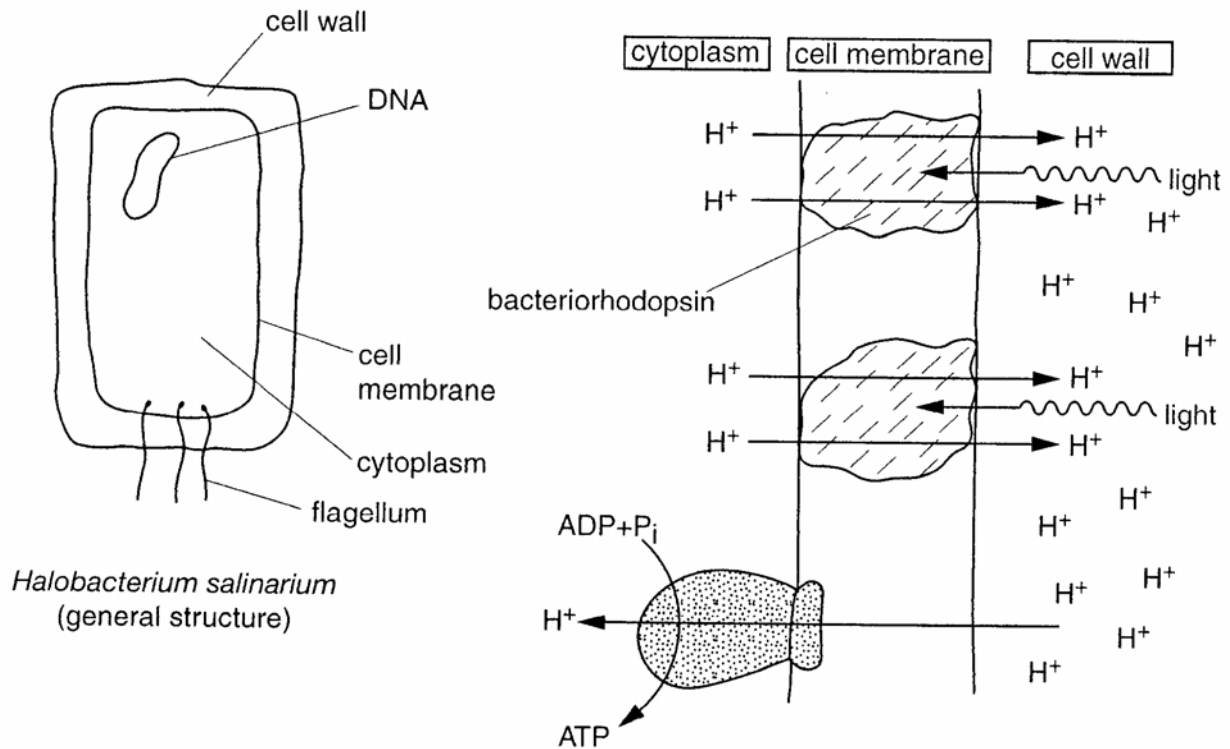


Fig. 7.2

With reference to Fig. 7.2, explain how *H. salinarium* synthesizes ATP.

[4]

[Total: 9m]

- 8 (a) The diagram below represents the various birth weights of new-born calves in a wild population of deer in the African grasslands. The line graph on the diagram represents mortality in relation to birth weight.

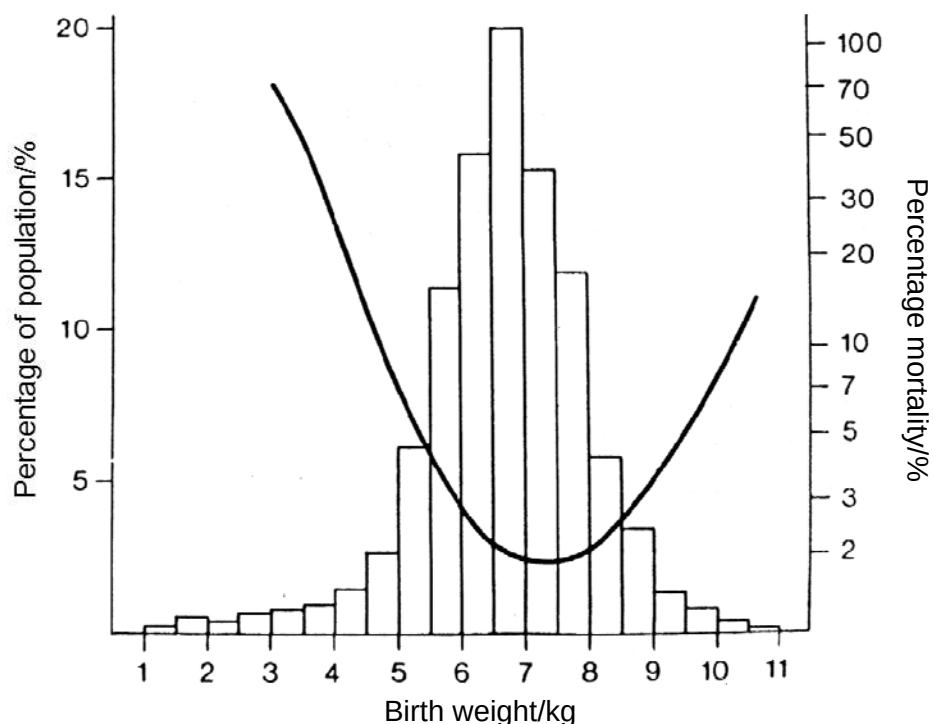


Fig. 8.1

- (i) Using the information provided in Fig. 8.1, account for the type of selection acting on birth weight of new-born deer calves. [4]

- (ii) Birth weight of new-born deer calves is an example of continuous variation. Explain why there is a variation of birth weights in the population of deer. [3]

- (iii) Suggest why percentage of mortality is higher on both ends of the range of birth weights of new-born deer calves. [2]

- (b) The deer is an artiodactyl. Artiodactyls are cloven-hooved mammals; they have hooves split into two toes and their weight is borne equally by the two toes. Fig. 8.2 shows a simplified phylogenetic tree of commonly-known artiodactyls and the skeletal anatomy of their hooves.

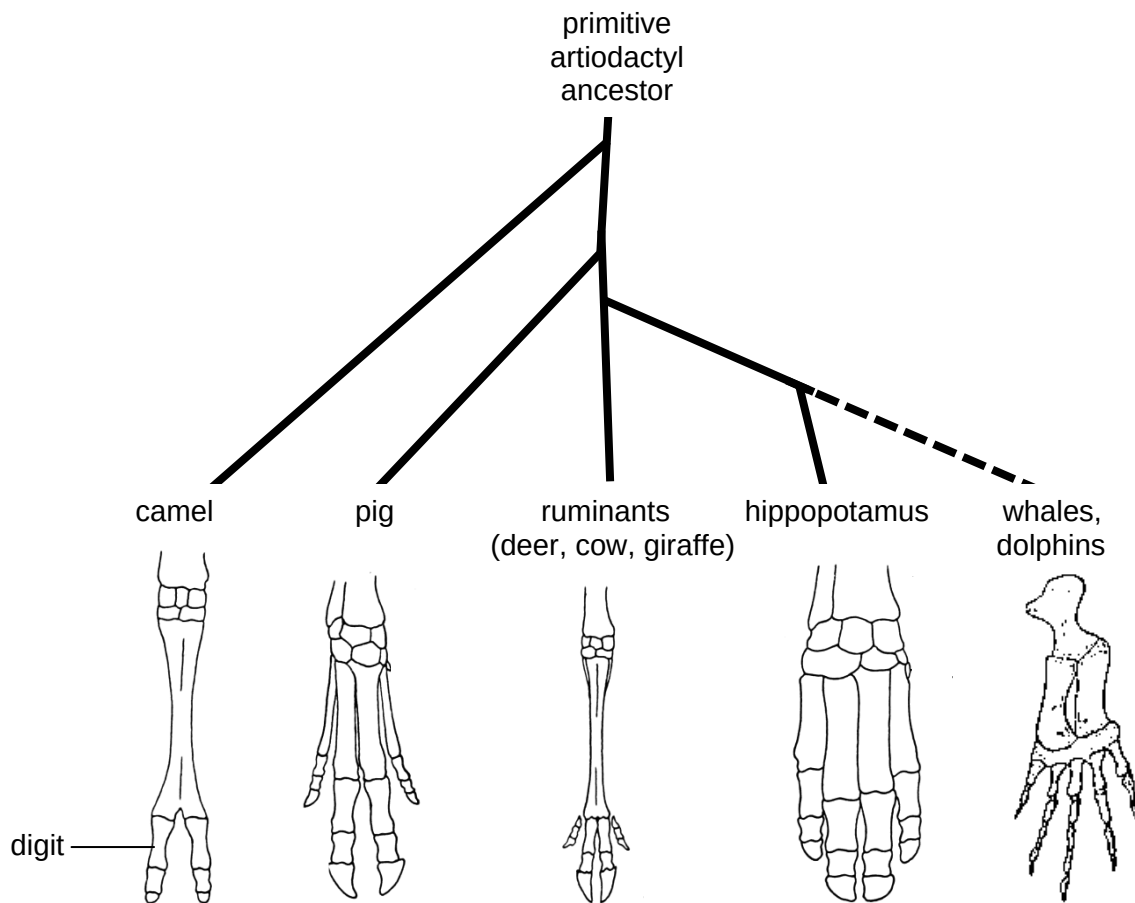


Fig. 8.2

- (i) On Fig. 8.2, use a 'X' to mark where the common ancestor of pigs, ruminants and hippopotamuses could have been positioned on the phylogenetic tree. [1]
- (ii) Recent studies have proposed that the closest relative of aquatic mammals such as whales and dolphins are the hippopotamuses (as indicated by the dotted line in Fig. 8.2). Besides using comparative anatomy, suggest and briefly explain the basis of one other possible approach scientists could have taken to arrive at this proposal. [2]

[Total: 12m]

Section BAnswer **EITHER 9 OR 10.**

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.**EITHER**

- 9** (a) Outline the differences between facilitated diffusion and exocytosis. [6]
(b) Describe the process of translation. [8]
(c) Explain how most evolutionary changes is the result of genetic drift acting on neutral alleles. [6]

OR

- 10** (a) Outline the differences between prokaryotic and eukaryotic control of gene expression. [8]
(b) Explain how a deletion mutation in the gene for DNA polymerase III will prevent the occurrence of DNA replication. [6]
(c) Explain how biogeography supports the theory of evolution by natural selection. [6]

[Total: 20m]

- END OF PAPER -